Test Case Report — HealthCareDigitalTwinn

This report provides an industry-style test plan and test case matrix for the HealthCareDigitalTwinn project (ML pipeline + API + UI). It includes automated test results and manual acceptance checks.

1) Test scope and objectives

In scope

- ML pipeline behavior (training, artifacts, inference compatibility)
- API contract and key endpoints: `/health`, `/predict`, `/model-info`, `/analyze`, `/figures/{name}`
- Security hardening checks for file-serving endpoint (`/figures`)
- Frontend quality gates (lint/build) and end-to-end UI user journeys

Out of scope (explicitly)

- Clinical validation (this is an analytics demo, not a medical device)
- Real-world bias/fairness audits beyond basic dataset summary
- Load testing / performance benchmarks

2) Test environment (execution evidence)

- OS: macOS
- Python: 3.13.7 (project venv)
- FastAPI: 0.135.3
- scikit-learn: 1.8.0
- Node: v24.9.0
- npm: 11.6.0

3) Test strategy (levels)

- **Unit/Component tests:** pure-Python logic and schema validation (pytest)
- **API integration tests:** FastAPI `TestClient` exercises endpoints and contracts (pytest)
- **Frontend gates:** ESLint + production build (Vite)
- **Manual acceptance:** browser journeys covering the predictor and what-if simulator

3.1 Entry and exit criteria

Entry criteria

- Dependencies installed (Python venv + Node modules)
- For tests that require trained artifacts/figures: artifacts present under `artifacts/` and figures under `reports/figures/` (otherwise those specific tests SKIP)

Exit criteria

- Backend automated tests meet the run's pass criteria (no unexpected failures)
- Frontend quality gates pass (`npm run lint`, `npm run build`)
- Manual acceptance tests for critical journeys (predict + what-if) executed and PASS for submission

3.2 Test data and prerequisites

- Automated tests use synthetic data for ML smoke validation (no external downloads needed)

- For full UI/API demonstration with real data, ensure the processed dataset exists under `data/processed/` and run training to generate artifacts

4) Automated test execution

4.1 Backend automated tests (pytest)

Command

```
cd HealthCareDigitalTwinn
.venv/bin/python -m pytest -q
```

Result (local run evidence)

```
9 passed in 1.33s
```

4.2 Frontend quality gates (lint + build)

Command

```
cd frontend
npm run lint
npm run build
```

Result (local run evidence)

- ESLint: pass
- Vite build: pass

5) Test case matrix (industry-style)

Status definitions:

- **PASS:** executed and met expected results
- **SKIP:** intentionally skipped due to missing prerequisites (e.g., artifacts)
- **READY:** written and defined; to be executed manually in a browser

5.1 Backend / API test cases (automated)

ID	Area	Test case	Expected result	Status	
API-000	API	`GET /health`	HTTP 200; returns `{status: ok}`	PASS	`tests/test_api_endpoints.py`
API-001	API	`GET /model-info`	HTTP 200; returns `default_model`, `dataset`, `models[]`	PASS	`tests/test_api_endpoints.py`
API-002	API	`POST /analyze` smoke	HTTP 200; baseline prob in [0,1]; scenarios include deltas	PASS	`tests/test_api_endpoints.py`

API-003	API	`/analyze` multi-scenario length	Response scenarios length equals request length	PASS	`tests/test_api_endpoints.py`
API-004	API	`/predict` model switching (L1 if available)	`model=l2` and `model=l1` both return prob in [0,1]	PASS or SKIP*	`tests/test_api_endpoints.py`
API-005	Security	`/figures` rejects traversal	Names with `/`, `\\`, `..` or non-`.png` rejected	PASS	`tests/test_api_endpoints.py`
API-006	Interop	CORS preflight for Vite origin	OPTIONS returns allow-origin for `http://localhost:5174`	PASS	`tests/test_api_endpoints.py`
API-007	API	`/figures/{name}` smoke (known PNG)	If confusion matrix exists, it is served as an image	PASS or SKIP*	`tests/test_api_endpoints.py`

*PASS if required artifacts/figures exist; otherwise marked SKIP by the test.

5.2 ML pipeline test cases (automated)

ID	Area	Test case	Expected result	Status	Evidence
ML-001	Modeling	Train+predict smoke	Training returns metrics incl. `f1`; schema validates payload	PASS	`tests/test_smoke.py::test_train_and_pred`

5.3 Frontend test cases (quality gates + manual acceptance)

ID	Area	Test case	Expected result	Status	Evidence
UI-001	Frontend	ESLint	`npm run lint` succeeds	PASS	`frontend` scripts
UI-002	Frontend	Production build	`npm run build` succeeds	PASS	`frontend` scripts
UI-003	Journey	Predict risk	Inputs → Predict → label + probability render	READY	Browser run
UI-004	Journey	What-if curve updates	Run prediction → choose feature + Δ → curve renders; selected point updates	READY	Browser run

UI-005	Journey	Model switch L2↔L1	Toggle model; predict/what-if continue working	READY	Browser run
UI-006	Negative	Missing fields	UI blocks submit or shows a validation message	READY	Browser run
UI-007	Negative	Backend offline	UI shows API disconnected; no crashes	READY	Browser run

6) Execution notes / prerequisites

- Some tests intentionally **skip** if model artifacts are missing. To generate artifacts:

```
.venv/bin/python scripts/train.py --model logreg --tune --penalty l2 --threshold 0.4
--artifact-name logreg_pipeline
.venv/bin/python scripts/train.py --model logreg --tune --penalty l1 --threshold 0.4
--artifact-name logreg_pipeline_l1
```

7) Summary

- Backend automated suite: **PASS** (9/9)
- Frontend gates: **PASS** (lint + build)
- Manual UI acceptance tests: **READY** (defined and repeatable)